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3 May 1974
SUPERSEDING
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3 August 1971

MILITARY SPECIFICATION

PROJECTILE, 30MM, TARGET PRACTICE: 9252416

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one model of 30MM target practice Projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal
TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

Military
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-651 - Visual Inspection Standards for 20MM Ammunition and Components

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2.1 (Cont'd)

Military (Cont'd)

MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedure and Tables for Inspection by Attributes

DRAWINGS

U. S. Army Armament Command

FX11731796	- Cartridge, 30MM, Practice, XM639B1
D9252415	- Cartridge, 30MM, Practice, XM639E1
D9252416	- Projectile, 30MM, Target Practice: 9252416
D9252417	- Body and Rotating Band Assembly
D9252418	- Body
C8889445	- Windshield
B8889437	- Blank, Rotating Band for Projectile, 30MM
LI-9252416	- Index of Inspection Equipment List for: Projectile, 30MM, Target Practice: 9252416

PUBLICATIONS

U.S. Army Materiel Command

AMCR 715-505	- Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM B117-64 - Method of Test for Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

(Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.)

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3. REQUIREMENTS

3.1 General.-The projectile shall comply with the requirements of Drawing (Dwg.) D9252416 and with all requirements of the drawings and specifications shown thereon.

3.2 Manufacturing process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (see 6.3)

3.3 Banding.-The rotating band blank and the rotating band seat shall be clean at the time of banding. Shot or sandblasting shall not be used to clean the serrated band seat after knurling. The rotating band blank shall fill the band seat.

3.4 Windshield security.-The windshield shall be affixed to the body in a secure manner and shall comply with the unseating requirements of Dwg. D9252416.

3.5 Protective coating.

3.5.1 Phosphate coating.-The pretreatment process used shall comply with applicable requirements of Dwg. D9252417. The phosphate coating shall comply with the requirements of TT-C-490 for appearance and minimum weight, as applicable to the type of phosphate coating.

3.5.2 Paint coating.-The paint coating shall comply with applicable requirements of Dwg. D9252417. Paint adhesion shall comply with applicable requirements of TT-C-490.

3.5.3 Corrosion resistance.-The painted assembly shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

3.6 Metal parts security.-The projectiles, when assembled into test cartridges to yield a pressure of 115% of service pressure, shall not separate or break-up in the test for metal parts security.

3.7 Workmanship.-The requirements for workmanship are as specified by the applicable drawings, referenced specifications, and the following:

3.7.1 Processing defects.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, deep scratches, cuts, dents, unblended radii, metal flaws, surface defects, dirt, grease, rust, corrosion products, foreign matter, excessive surface roughness, and any other forms of substandard workmanship. Any cleaning method used shall not be injurious to any parts nor shall any parts be contaminated by the cleaning agents. All washing and stamping shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification and drawing where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample applicable to the approved process shall be submitted in accordance with contract requirements and shall consist of:

a. Inspection and non-firing tests:

125 Windshields
125 Bodies
125 Rotating bands
125 Body and rotating band assemblies
125 Projectiles
3 Standard panels

b. Firing tests:

1000 Projectiles

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. Identification shall be in accordance with MIL-STD-1168.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified by the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications will result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

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4.3.1.1 Lot.-A lot shall consist of projectiles produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

4.3.1.2 Lot identification.-Each projectile and each packed projectile lot shall be identified in accordance with applicable drawings and MIL-STD-1168, supplemented as directed by the procuring activity.

4.3.2 Examination.-One-hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Each part and each assembly found to be defective shall be rejected.

4.3.2.1 Sampling and acceptance criteria.-Except as otherwise indicated, sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. Defect characteristics may be grouped into a class and considered collectively, or defect characteristics may be considered individually to determine acceptability. When acceptability is determined on a class basis, in addition to the Acceptable Quality Level (AQL) assigned to the major class, an AQL shall be assigned to individual characteristics within the major class, limiting the number of defectives containing a particular major defect to less than the number of defectives permitted for the class. However, where sampling procedures of MIL-STD-1235 are used, acceptability shall not be determined on a class basis for major defects. In addition, where three or less defects appear in a classification, sampling plans applicable to a class shall not be used for major defects. (see 6.4)

4.3.2.2 AQL's.-The following AQL's are assigned to major and minor defects:

a. Individual:

Major	0.25%
Minor	0.40%

b. Class basis:

Major	1.50%
Minor	2.50%

c. Individual defects associated with the class basis:

Major	0.40%

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4.3.3 Classification of defects.-1/ The classification of defects shall be as specified herein. (see 6.5)

4.3.3.1 Windshield, Drawing D8889445, covering a detail of Drawing D9252416.

Categories and Defects	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Location large basic diameter	Gage
102. Location small basic diameter	Gage
MINOR	
201. Length, overall	Gage
202. Thickness, wall, at open end	Gage
203. Evidence of poor workmanship (See 3.3)	Visual

4.3.3.2 Blank, Rotating Band, Drawing B8889437, covering a detail of Drawing D9252417.

Categories and Defects	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Hardness, max.	Rockwell Hardness Tester
MINOR	
None defined	

4.3.3.3 Body, Drawing D9252418, covering a detail of Drawing D9252417.

Categories and Defects	Method of Inspection
CRITICAL	
None defined	

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4.3.3.3 (Cont'd)

MAJOR		
101.	Length, open end to bourrelet shoulder	Gage
102.	Diameter, bourrelet	Gage
103.	Runout, bourrelet with body diameter	Gage
104.	Runout, open end with body diameter	Gage
105.	Diameter, body	Gage
106.	Location, basic diameter, closed end	Gage
107.	Runout, closed end with body diameter	Gage
108.	Location, large basic diameter, open end	Gage
109.	Location, small basic diameter, open end	Gage
110.	Depth, groove forward of bourrelet	Gage
MINOR		
201.	Length, overall	Gage
202.	Depth, cavity	Gage
203.	Runout, cavity with body diameter	Gage
204.	Diameter, cavity	Gage
205.	Length, rear of rotating band groove to open end	Gage
206.	Diameter, middle of band groove	Gage
207.	Depth, crimp grooves, rear of band groove	Gage
208.	Evidence of poor workmanship (See 3.3)	Visual

4.3.3.4 Body and Rotating Band Assembly, Drawing D9252417, covering a detail of Drawing D9252416.

Categories and Defects		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Length, rear of rotating band to open end	Gage
102.	Diameter, bourrelet, max.	Gage
103.	Diameter, large rotating band	Gage

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4.3.3.4 (Cont'd)

MAJOR (Cont'd)

104.	Diameter, small rotating band	Gage
105.	Runout, large rotating band with body diameter	Gage
106.	Runout, small rotating band with body diameter	Gage
107.	Length, rear of large band to angle of small band	Gage

MINOR

201.	Evidence of poor workmanship (See 3.3)	Visual
202.	Diameter between rotating bands	Gage
203.	Width, rear rotating band	Gage
204.	Width between rotating bands	Gage

4.3.3.5 Projectile, 30MM, Target Practice: D9252416.

Categories and Defects	Method of Inspection

CRITICAL

None defined

MAJOR

101.	Runout, windshield with body diameter	Gage
102.	Length, overall, max.	Gage

MINOR

201.	Marking missing, illegible or improper	Visual
202.	Evidence of poor workmanship (See 3.3)	Visual

1/ MIL-STD-651 shall be used as a guide in defining and evaluating visual defects.

4.3.4 Tests.-The tests listed in Table I shall be conducted on each lot in accordance with the applicable methods and procedures prescribed in 4.4. For acceptance, the results of each test shall comply with the applicable requirement(s).

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4.3.4.1 Test samples.-The sample sizes for the various tests shall be as specified in Table I.

TABLE I

Test	Sample Size		Item	Req. Para.
Band tightness	1/c		Body and Rotating Band Assemblies	3.3
Windshield security	2/	80	Projectiles, painted	3.4
Phosphast coating (appearance & weight)	3/	3	Standard panels (See TT-C-490)	3.5.1
Paint adhesion	4/	13	Projectiles, painted	3.5.2
Corrosion resistance	5/	32	Projectiles, painted	3.5.3
Metal parts security	6/	250	Projectiles, painted	3.6

1/ Band tightness.

a. Machine qualification.-Prior to submission of first article, a sample of fifty (50) body assemblies will be examined for existence of a gap between the band seat of the body and the inside diameter of the rotating band. If a gap if found in one (1) assembly, the sample will be rejected and a new sample of fifty (50) submitted. The inspection may be performed by any qualified government representative and a copy of the data forwarded to the responsible technical agency.

b. First article.-After acceptance of the machine qualification sample, a sample of fifty (50) body assemblies will be submitted for examination of existence of a gap between the band seat of the body and the inside diameter of the rotating band. If a gap is found in one assembly, the sample will be rejected and a new sample of fifty (50) submitted.

c. Production.-After acceptance of the first article, for the first week of the production run a sample of five (5) body assemblies will be taken from each four (4) hours' production, or fraction thereof, per machine. The sample will be selected in such a manner as to be representative of the entire four (4) hours' production. These assemblies will be examined for evidence of a gap between the band seat and the rotating band. If no gap if found, then commencing with the second week's production and continuing thereafter, the sample may be reduced to one (1) assembly from each two (2) hours' production, or fraction thereof, per machine.

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1/c (cont'd)

If, at any time, a gap is found in one (1) assembly, then a random sample of ten (10) assemblies taken from the units produced since the last inspection will be examined for evidence of gap. If none is found, the production from which the sample was taken may be included in regular production. If a gap is found in one (1) assembly, then the production from which the sample was taken may not be included in regular production.

Whenever a defect is found, sampling five (5) body assemblies every four hours' production, or fraction thereof, per machine will be employed commencing with the next sample taked.

2/ If two or more units of the sample fail to comply with the requirements for windshield security, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples two or more units fail to comply with the requirement for windshield security, the lot shall be rejected.

3/ Failure of the phosphate coating on the standard panels to comply with the applicable requirements shall be cause for rejection of all projectiles phosphated since the preceding test.

4/ If two or more units of the sample fail to comply with the requirement for paint adhesion, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples two or more units fail to comply with the requirement for paint adhesion, the lot shall be rejected.

5/ If two or more units of the sample fail to comply with the corrosion resistance requirements, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

6/ The occurrence of one or more of the following casualties, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s) shall be cause for rejection of the lot, except as otherwise provided:

a. Body breakup.

b. Loss of nose.

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6/ (Cont'd)

c. Complete rotating band separation. This is defined as detachment of all of the band having band seat knurling imprint.

d. Partial rotating band separation. This is defined as detachment of part of the band having band seat knurling imprint.

e. The following acceptance criteria shall apply to thrown band fragments having no evidence of band seat knurling imprint:

(1) The lot will be accepted if two or less assemblies throw band fragments having length not greater than 0.500 inch.

(2) The lot will be rejected if three or more assemblies throw band fragments having length not reater than 0.500 inch.

(3) The lot will be rejected if any assembly throws a band fragment having length greater 0.500 inch (indicative of deep-seated band lamination).

(4) The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering.

4.3.5 Packaging, packing and marking inspection.-Inspection for packaging, packing and marking shall comply with applicable requirements of MIL-STD-1169.

4.3.6 Inspection equipment.-Index of inspection equipment list LI9252416 identifies the applicable inspection lists indicating the equipment required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable EL (reference is made to Inspection Equipment List for definition of coding format).

4.4 Test methods and procedures.

4.4.1 Band tightness (destructive test).-The diameter of the groove in the rotating band shall be measured on three (3) locations, 60 degrees apart, at a point midway between the two large band diameters. This comprises three (3) diametrical measurements. These points of measurement shall be marked on the band and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The band shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 6 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the band seat diameters the sum of the corresponding two (2) thicknesses of the band. The results thus obtained shall then be subtracted from the previously determined diameters over the band.

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4.4.2 Security of windshield on body.-The projectile shall be placed in a suitable fixture and the minimum unseating force specified on Drawing D9252416 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.3 Phosphate coating.-The method of test shall be as prescribed by TT-C-490 for the type of coating used (see Drawing D9252417). The prescribed panels shall be processed, with the projectiles to be represented, at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

4.4.4 Paint adhesion.- The method of test shall be as prescribed by TT-C-490. The test shall be made on the painted bourrelet of each unit of the sample forward of the rotating band. Visible exposure of underlying phosphate coating of ferrous metal of any unit of the sample as a result of the test, shall be recorded as a defect.

4.4.5 Corrosion resistance.-The method of test shall be as prescribed in American Society for Testing and Materials (ASTM), ASTM B117. The paint shall not be disturbed, however, the rotating band and unpainted surfaces of the sample projectiles may be masked out by a wax coating or other suitable material. The salt spray solution and exposure time shall be as specified in 3.5.3.

4.4.6 Metal parts security.-The method of test will be as prescribed by AMCR-715-505, Volume 8, as applicable. The test samples will be assembled into cartridges to comply with Drawing D9252415 or Drawing FX11731796, as applicable, except that sufficient propellant will be used to yeild a pressure of 32,200 PSI, which is 115% of the service pressure. The test cartridges, at ambient temperature, will be fired in the XM140E3, 30MM weapon in bursts of 25 shots. The gun barrel will be at ambient temperature for the firing of each burst. The cyclic rate of fire will be recorded for informational purposes.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.

5.1.1 Level C.-The projectiles shall be packed and marked for shipment in accordance with MIL-STD-1169. The projectiles shall be packed in accordance with 3.1.4 of MIL-STD-1169 in celled shipping containers which contain one hundred (100) projectiles maximum. The shipping containers shall be palletized for shipment in accordance with 3.6.2 of MIL-STD-1169.

5.2 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of lots in transit.

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6. NOTES

6.1 Ordering data.-Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number, and date of this specification.

6.1.2 Place of inspection, if not place of manufacture.

6.1.3 Packing and marking instructions. (see 5.1).

6.1.4 Provision for shipment and testing of production lot samples for metal parts security. (see 4.4.6)

6.1.5 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.6 Requirement for the contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirement.

6.2 First article sample.-Where an initial production sample is required by the contract, the contracting officer will notify the test facility when to proceed with the metal parts security test (see 4.2). Additional projectiles may be required by the test facility.

6.3 Process deviation.-A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item of contract.

6.4 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might rwsult from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.5 Intermediate point inspection.-The classification of defects indentifies the defect characteristics (among other things) for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality in the end item.

Custodian:

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Preparing activity:

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Project No. 1305-A758

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